

CHARTERED INSTITUTE OF TECHNOLOGY, ABU ROAD

B.Tech. II Sem I Internal Examination, 2023-24

SUBJECT CODE: BEE1st Year / 2nd Sem

10:00 AM TO 12:00 PM

DATE: .05.2024

STAFF NAME:-jaydeep kumar

Maximum Marks: 60

Minimum Marks :

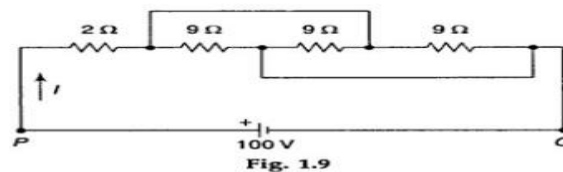
SECTION -A**Short Answer Questions – (5 Questions, each of 2 marks) (2X5=10)**

1. What is KVL or KCL?
2. What is current?
3. What is potential difference?
4. Write Representation of sinusoidal waveform?
5. Write a unit , resistance ,inductions, capacitance ?

SECTION-B**Medium Answer based Questions – (Any Three, each of 8 marks) (3 X 8=24)**

1. Write a thevenins theorem ?
- 2.

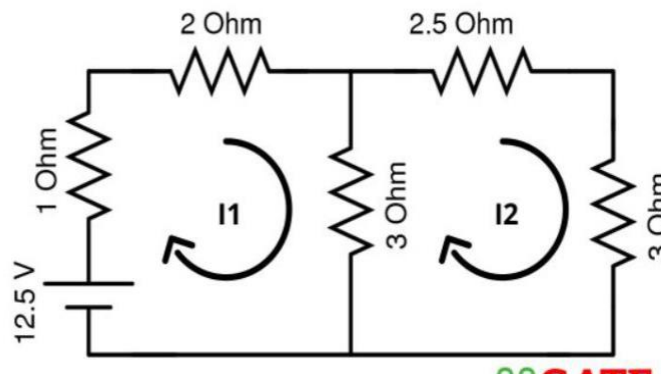
Fundamentals 31
 10. Find the power consumed by the combination of resistances when connected to a voltage source of 100 V at terminals P-Q (Fig. 1.9).
 (Ans: 3.125 kW)



[Hint: Fig. 1.9 can be reduced as shown below:

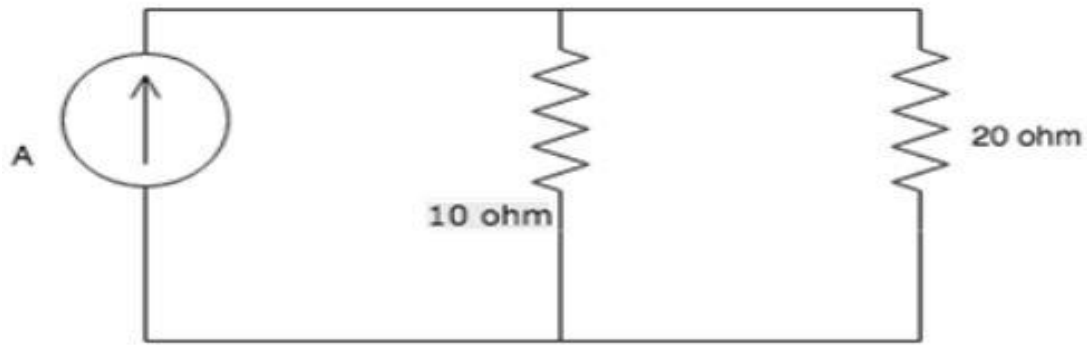
3. .

- 3) By using Kirchhoff's voltage law (KVL) / Mesh analysis find the current flowing through all resistances.



4. .THE Given current 5aA

Calculate the current across the 20 ohm resistor.



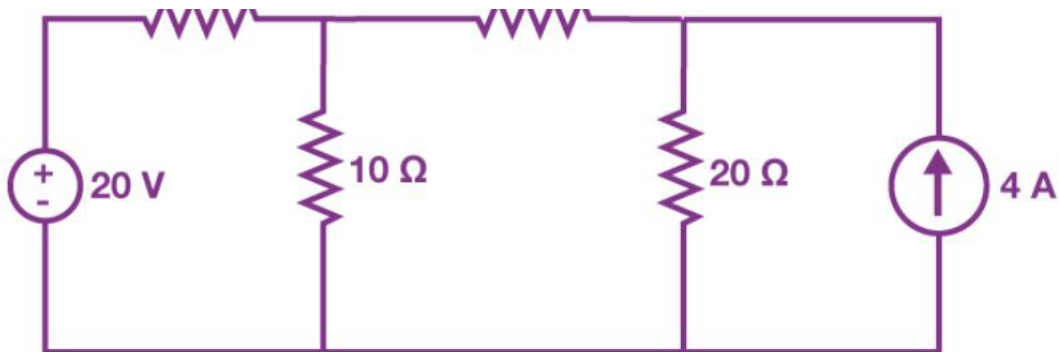
20A

5. difference between Mesh and loop analysis ?

SECTION-C

Long Answer based Questions – (Any two, each of 13 marks) (2 X 13=26)

1. Star To Delta Transformation Or Delta To Star Transformation drive expression
2. Drive RMS value of current and average value OF CURRENT full wave ?
3. Explain Superposition theorem ? solve by superposition theorm $r_1=5$ and $r_2=10\text{ohms}$ calculate the cureent through 20 ohms ?



4. Write a Property of alternating quantity ?